



Merit Education Group

Unit 1 Overview

Variables and Expressions — Intermediate Algebra · Merit Education Group™

What You Will Learn

Welcome to Unit 1: Expressions (Variables and Expressions). This unit spans 20 days — 17 lessons and 3 assessment/flex days — and targets the 2022 Minnesota algebra standards: 9.3.5.1, 9.3.5.3, 9.3.5.5, 9.3.5.7, 9.3.5.8. Each lesson begins with a whiteboard bell ringer, builds new skills through guided and independent practice, and closes with an exit ticket.

Lesson Map

Lesson	Title	Theme
L1	What Is a Variable?	Minnesota State Fair
L2	Translating Verbal Phrases	Lake Superior & Duluth
L3	GEMDAS Basics	Juicy Lucy Diner
L4	GEMDAS — Advanced	Boundary Waters Canoe Area
L5	Scientific Notation	Itasca State Park
L6	Evaluating — One Variable	Minnehaha Falls
L7	Evaluating — Multiple Variables	Ice Fishing on Minnesota Lakes
L8	Real-World Formulas	Mill City Museum & Stone Arch Bridge
L9	Identifying Terms, Coefficients & Constants	Tater Tot Hotdish
L10	Combining Like Terms	Itasca State Park
L11	The Distributive Property	Paul Bunyan & Babe the Blue Ox
L12	Dist. Property + Combining Like Terms	State Capitol
L13	Equivalent Expressions	North Shore
L14	Writing Equivalent Expressions (GCF)	Generic Shopping Mall
L15	Perimeter, Area & Algebraic Expressions	Generic Amusement Park
L16	Translating Real-World Scenarios	Mississippi River
L17	Financial Literacy: Budgeting	Sports Stadium

Assessment Schedule

Day	Assessment	Covers
7	Quiz 1	Quiz 1 (covers L1–L5)
17	Quiz 2	Quiz 2 (covers L6–L15)

Day	Assessment	Covers
20	Summative	End-of-Unit Test (covers L1–L17)

Unit 1 Rubric: How Grades Are Earned

Grades are earned on a locked 4-level scale: Emerging → Developing → Proficient → Superlative. Proficient is the passing level; Superlative earns an A.

Benchmark	Emerging	Developing	Proficient	Superlative
9.3.5.3	Recognizes but cannot yet apply number systems.	Applies number systems with support or on routine problems only.	Correctly applies number systems to solve problems and explain reasoning.	Independently applies number systems to novel problems and justifies the most efficient approach.
9.3.5.5	Recognizes but cannot yet apply cost, discounts, taxes & order of operations.	Applies cost, discounts, taxes & order of operations with support or on routine problems only.	Correctly applies cost, discounts, taxes & order of operations to solve problems and explain reasoning.	Independently applies cost, discounts, taxes & order of operations to novel problems and justifies the most efficient approach.
9.3.5.7	Recognizes but cannot yet apply equivalent forms & rearranging formulas.	Applies equivalent forms & rearranging formulas with support or on routine problems only.	Correctly applies equivalent forms & rearranging formulas to solve problems and explain reasoning.	Independently applies equivalent forms & rearranging formulas to novel problems and justifies the most efficient approach.
9.3.5.8	Recognizes but cannot yet apply writing expressions in multiple ways.	Applies writing expressions in multiple ways with support or on routine problems only.	Correctly applies writing expressions in multiple ways to solve problems and explain reasoning.	Independently applies writing expressions in multiple ways to novel problems and justifies the most efficient approach.

Proficient is the passing level for this unit; Superlative earns an A. Grades are earned across the benchmarks above — see your teacher for details.

End of U1_Overview — generated by regenerate_overview.py from the canonical Unit 1 spec.